CSC1042 Computer Network II

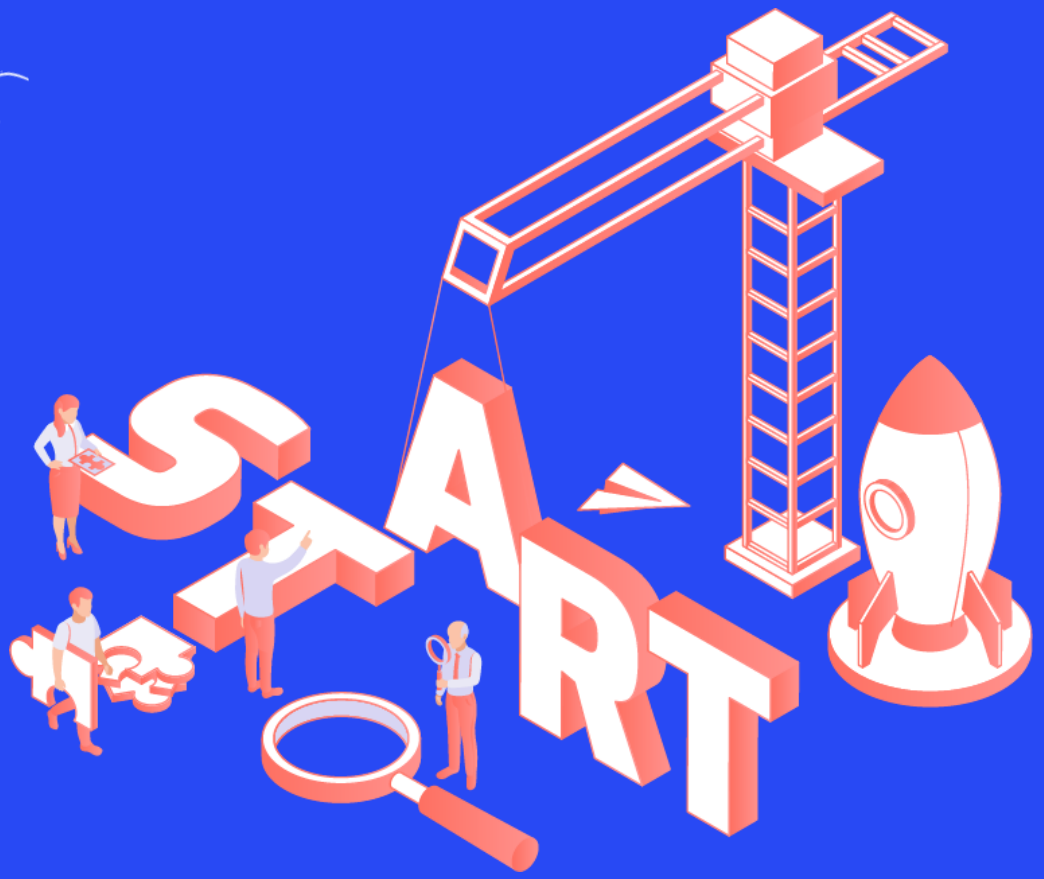
Dynamic Host Configuration Protocol

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Learning outcomes

- Describe the service provided by DHCP
- Explain how DCHP works
- Describe DHCP's address allocation methods



Dynamic Host Configuration Protocol - DHCP

- A protocol used to assign IP addresses to hosts in a network
- It introduces the notion of “leasing IP addresses”
- A host might have a different IP address every time it connects to the network → thus, the term “dynamic”
- IP addresses are leased to hosts from a pool of IP addresses
- Default lease time is 1 day, but can be configured by the network admin

DHCP – how it works?

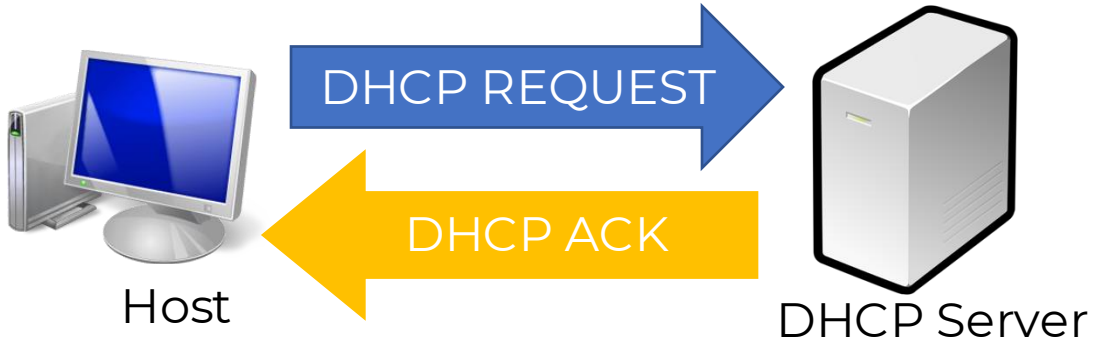
- When a host wants to connects to a network, it sends a DHCP DISCOVER message
  - It contains the host name and MAC address



- The server replies by sending DHCP OFFER message
  - It contains the IP address, network mask, the default gateway address
  - Before sending DHCP OFFER, the server make sure that IP address is not used

DHCP – How it works?

- Once DCHP OFFER received, the host sends DHCP REQUEST message
  - It contained the assigned IP address and the server's IP address



- The server then replies by sending DHCP ACK message
  - This finalizes the IP address lease
- The host will need to renew the lease before it expires if they wish to use the same IP address by sending DHCP REQUEST

DHCP – Address allocation methods

- Dynamic allocation
  - Hosts are given an IP address from a pool of addresses
- Automatic allocation
  - Hosts are the given the same previously offered IP address if not used (often combined with Dynamic)
- Static allocation
  - Hosts are allocated a reserved IP address for their MAC address
  - MAC addresses should be entered into the DHCP Server allocation table

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THANK YOU